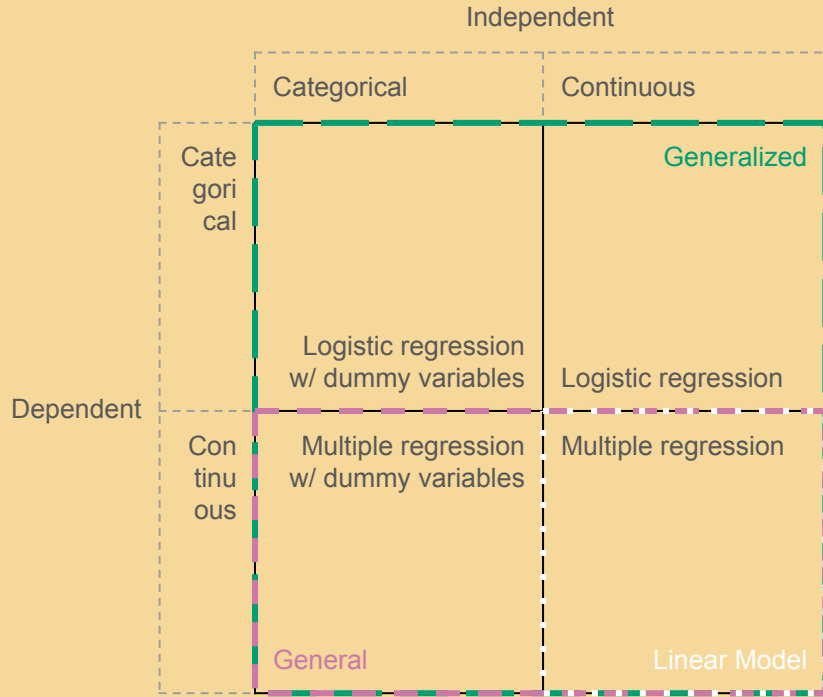


Multilevel & Longitudinal Analysis

for dependent observations



Recap





Today

Topics

- 1 | Statistical reasoning with GLM
- 2 | Multiple linear regression
- 3 | Dummy-variable regression
- 4 | Bayesian statistics
- 5 | Logistic regression
- 6 | Multilevel and longitudinal analysis
- 7 | Statistics superpowers

Examples



Depression (longitudinal)



Danceability (cross-sectional)

Hierarchical Analysis

for dependent observations



Wisdom of the crowd?



How tall am I?



How old am I?



[Social desirability bias](#)

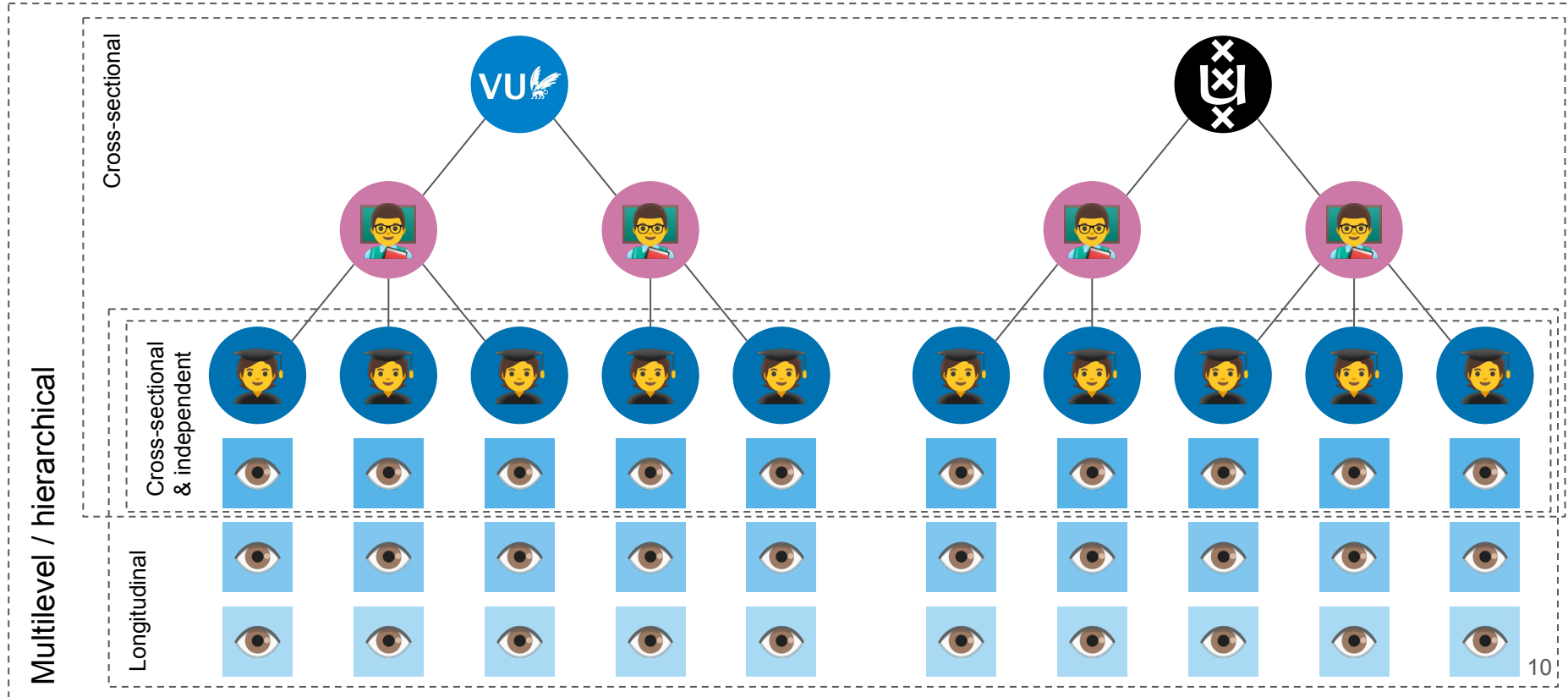


[Anchoring effect](#)



[Veritasium](#) on dependent events (first five minutes), Markov chains, atomic bombs, and LLMs

Dependent observations



Music

Danceability

“Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.”

— [Spotify](#)

How are the danceability and popularity of tracks related?

Popularity

“The popularity of a track is a value between 0 and 100, with 100 being the most popular. The popularity is calculated by algorithm and is based, in the most part, on the total number of plays the track has had and how recent those plays are.”

— [Spotify](#)

danceability 

popularity 

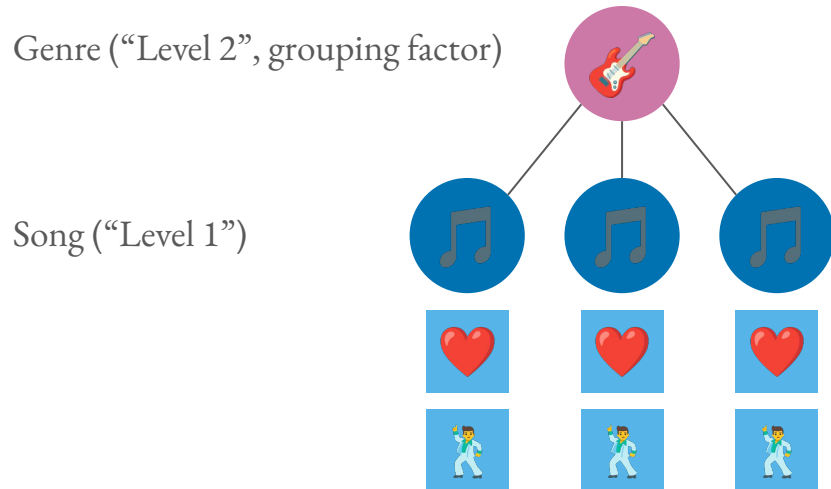
Dependent observations

That only requires ordinary regression. But what if songs within genres are more alike in popularity than songs across genres, beyond what danceability explains? 🦸

🦸 You mean because there are fewer of us metal fans than you hip hop fans?

Exactly. Moreover, what if the relationship between danceability and popularity also depends on genre? 🦸

🦸 Are you saying I don't like danceable songs and you do? Watch my moves!



Dependent observations

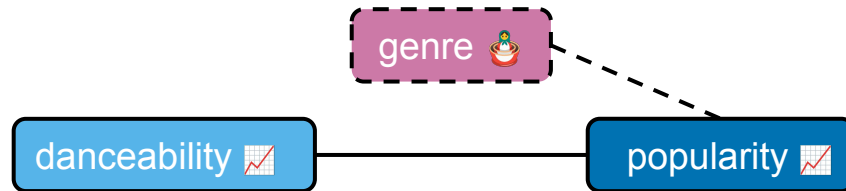
That only requires ordinary regression. But what if songs within genres are more alike in popularity than songs across genres, beyond what danceability explains? 🦸

🦸 You mean because there are fewer of us metal fans than you hip hop fans?

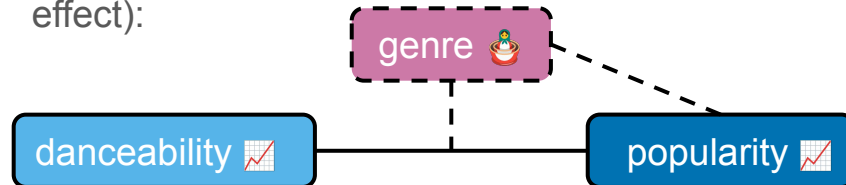
Exactly. Moreover, what if the relationship between danceability and popularity also depends on genre? 🦸

🦸 Are you saying I don't like danceable songs and you do? Watch my moves!

Popularity depends on genre (level effect):

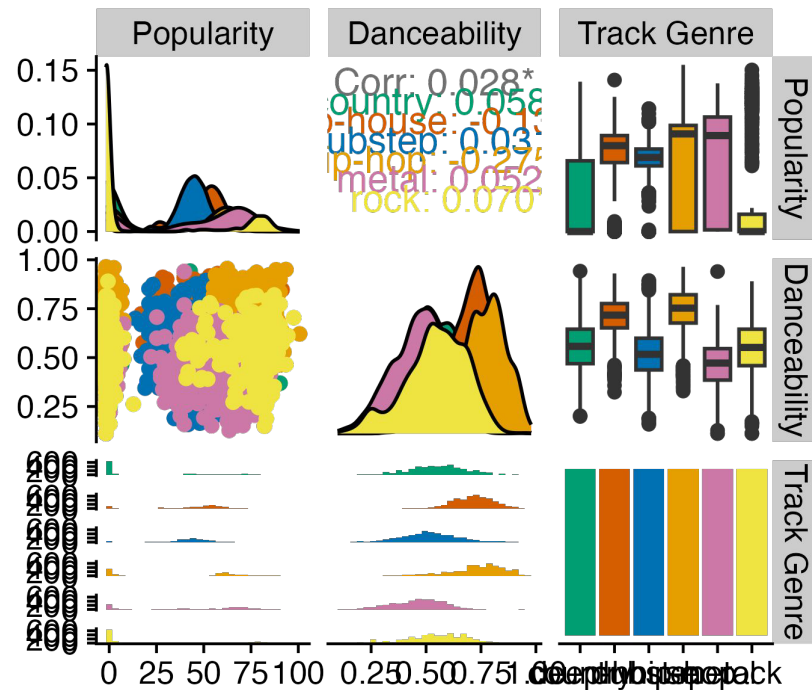


And the relationship between danceability and popularity depends on genre (relationship effect):



Data

```
library("moderndive")
data("spotify_by_genre")
ggpairs(spotify_by_genre,
        mapping = aes(color = track_genre))
```



Simple regression | Complete pooling



Musical genres imprison the creativity of humankind.

We should model it without such artificial limitations:

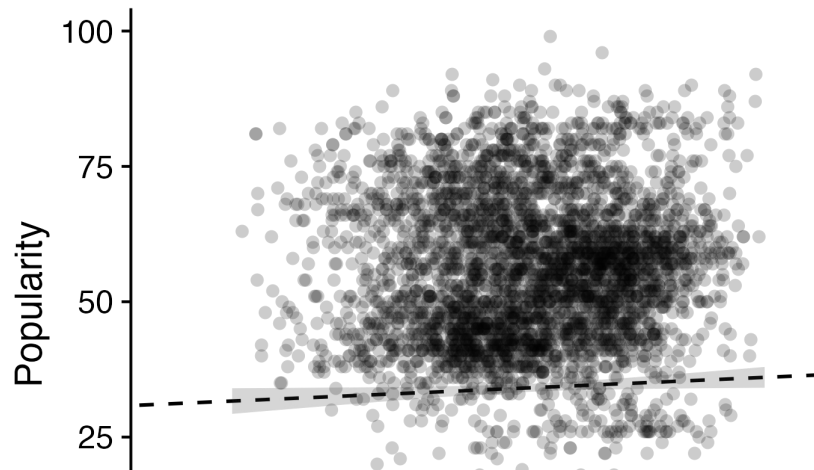
```
Call:
lm(formula = mod, data = spotify_by_genre)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-36.034 -33.353   8.325  24.205  64.727
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   31.084     1.463   21.253  <2e-16 ***
danceability    5.134     2.404    2.136   0.0328 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 29.07 on 5998 degrees of freedom
Multiple R-squared:  0.0007598, Adjusted R-squared:  0.0005932
F-statistic: 4.561 on 1 and 5998 DF, p-value: 0.03275
```

```
mod <- popularity ~ danceability
fit <- lm(mod, data = spotify_by_genre)
```



*Way to go, that's what we call complete pooling, or “naive regression”. Naivety may fit you well, but it's **bad for the fit of the model**. What if this relation differs across genres? 🦸*

Moderation | No pooling

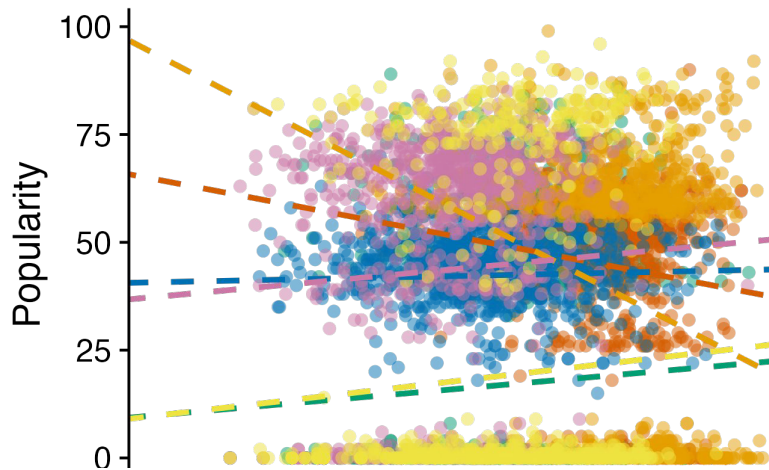


You win, I'll fit a separate model for each genre. Or, equivalently, an interaction model where the slopes are allowed to vary:

track_genre	danceability.trend	SE	df	lower.CL	upper.CL
country	12.7	6.71	5988	-0.455	25.8
deep-house	-27.6	9.02	5988	-45.275	-9.9
dubstep	3.0	6.59	5988	-9.921	15.9
hip-hop	-75.1	7.01	5988	-88.865	-61.4
metal	13.5	7.20	5988	-0.619	27.6
rock	16.7	6.15	5988	4.653	28.8

Confidence level used: 0.95

```
mod <- popularity ~ danceability * track_genre
fit <- lm(mod, data = spotify_by_genre)
emmeans(fit, "track_genre", at = list(danceability = 0)) # genre-specific intercepts
emtrends(fit, "track_genre", var = "danceability")
# genre-specific slopes
```



*Clever, but it's **an answer to a different question** and **it comes at the cost of statistical power**. While the relationships are permitted to differ, they can no longer benefit from what they share. What's in between? 🤖*

Mixed effects | Partial pooling

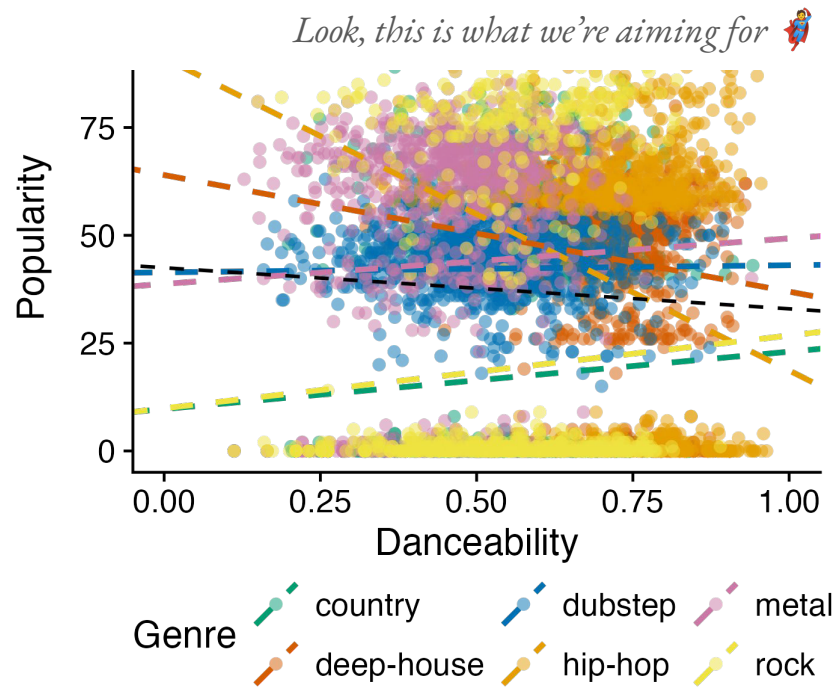
$$\begin{aligned}Y_{i,j} &= \beta_0 + \beta_1 X_{i,j} + u_i + v_i X_{i,j} + \varepsilon_{i,j} \\ &= (\beta_0 + u_i) + (\beta_1 + v_i) X_{i,j} + \varepsilon_{i,j}\end{aligned}$$

$$u_i \sim \text{Normal}(0, \sigma_u)$$

$$v_i \sim \text{Normal}(0, \sigma_v).$$

Fixed effect models the average relationship.

Random effects model the genres' deviations from the average, coming from a group-level normal distribution.



Random intercepts | Model thinking

How are the danceability and popularity of tracks related?

genre 🍷

danceability 📈

popularity 📈

$$\begin{aligned} Y_{ij} &= \beta_0 + \beta_1 X_{ij} + u_i + \varepsilon_{ij} \\ &= (\beta_0 + u_i) + \beta_1 \times X_{ij} + \varepsilon_{ij} \end{aligned}$$

```
mod <- popularity ~ danceability + (1 |  
  track_genre)
```

Fixed effects: `intercept + predictors`

- ❑ Intercept
- ❑ Danceability

Random effects: `(what is nested | in what)`

- ❑ Intercept



Book shows incorrect regression equation

Random intercepts | Results

```
fit <- lmerTest::lmer(mod, data = spotify_by_genre)
summary(fit)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: popularity ~ danceability + (1 | track_genre)
Data: spotify_by_genre

REML criterion at convergence: 56450.8

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.72633	-0.68266	-0.00086	0.74000	2.92395

Random effects:

Groups	Name	Variance	Std.Dev.
track_genre	(Intercept)	164.8	12.84
Residual		711.2	26.67

Number of obs: 6000, groups: track_genre, 6

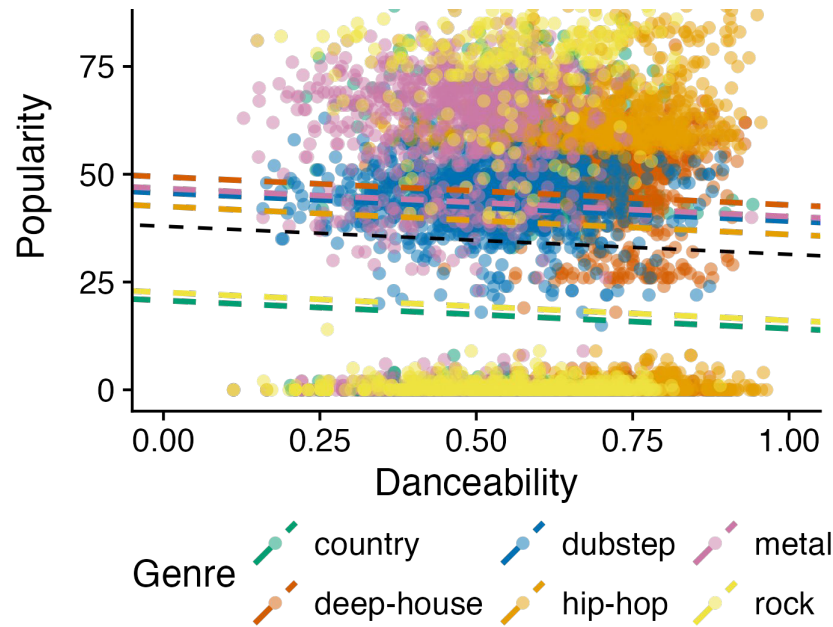
Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	37.936	5.517	6.080	6.877	0.00044 ***
danceability	-6.519	2.870	5964.492	-2.271	0.02319 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
(Intr)
danceability -0.306

This resembles a repeated-measures ANOVA 🤖



Random intercepts | Intraclass correlation coefficient

```
icc(fit)
```

```
# Intraclass Correlation Coefficient
```

```
Adjusted ICC: 0.188
```

```
Unadjusted ICC: 0.188
```

Proportion of variance 'explained' by the grouping defined by the random intercepts

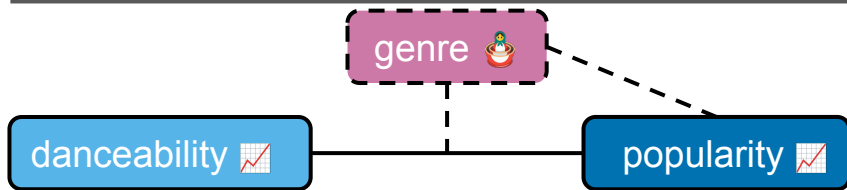
ICC = 1 (grouping 'explains' the data perfectly)

ICC = 0 (grouping doesn't add to random grouping)

What do we think of the ICC?

Random intercepts & slopes | Model thinking

How are the danceability and popularity of tracks related?



$$\begin{aligned} Y_{i,j} &= \beta_0 + \beta_1 X_{i,j} + u_i + v_i X_{i,j} + \varepsilon_{i,j} \\ &= (\beta_0 + u_i) + (\beta_1 + v_i) X_{i,j} + \varepsilon_{i,j} \end{aligned}$$

```
mod <- popularity ~ danceability + (1 +  
  danceability | track_genre)
```

Fixed effects: `intercept + predictors`

- ❑ Intercept
- ❑ Danceability slope

Random effects: `(what is nested | in what)`

- ❑ Intercept
- ❑ Danceability slope

Random intercepts & slopes | Results

```
fit <- lmerTest::lmer(mod, data = spotify_by_genre)
summary(fit)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: mod
Data: spotify_by_genre

REML criterion at convergence: 56338.6

Scaled residuals:
    Min       1Q   Median       3Q      Max
-2.35235 -0.69640 -0.01326  0.67670  2.81848

Random effects:
Groups: Name Variance Std.Dev. Corr
track_genre (Intercept) 1014.8  31.86
           danceability 1250.9  35.37   -0.94
Residual          696.3  26.39
Number of obs: 6000, groups: track_genre, 6

Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)   42.477     13.131  4.971   3.235  0.0233 *
danceability  -9.522     14.731  5.014  -0.646  0.5464
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
(Intr)
danceabilty -0.936
```

Fixed effect

Intercept: average popularity where danceability is 0, before genre offset

Slope: average decrease in popularity when danceability increases by 1, before genre offset (divide by 10 for danceability increase by .1) > not significant

Random effects

Intercept + slope: fluctuations of individual coefficients (for genres), summarized by standard deviations (mean = 0)

Evaluation | Confidence intervals

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
track_genre	(Intercept)	1014.8	31.86	
	danceability	1250.9	35.37	-0.94
Residual		696.3	26.39	

Number of obs: 6000, groups: track_genre, 6

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	42.477	13.131	4.971	3.235	0.0233 *
danceability	-9.522	14.731	5.014	-0.646	0.5464

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
confint(fit, method = "boot", level = .95, nsim = 1000)
```

	2.5 %	97.5 %
.sig01	11.5578612	52.5127103
.sig02	-0.9987584	-0.6455147
.sig03	13.4642573	56.9573436
.sigma	25.9508460	26.8382483
(Intercept)	15.7986835	69.6890800
danceability	-39.5344842	19.0832627

- ❑ Does it make sense to include the random intercept and random slope?
- ❑ Is there an average effect of danceability?

Evaluation | Fixed & random effects

```
fixef(fit); ranef(fit); coef(fit)
```

```
$`(Intercept)`  
[1] 42.47747  
  
$danceability  
[1] -9.522212  
  
$track_genre  
      (Intercept) danceability  
country    -32.753747      22.81309  
deep-house  21.501598     -17.51555  
dubstep    -1.064435      11.17599  
hip-hop    48.654123     -62.94547  
metal     -3.749130      20.08089  
rock     -32.588409      26.39105  
  
$track_genre  
      (Intercept) danceability  
country    9.723725      13.290874  
deep-house 63.979070     -27.037764  
dubstep    41.413037      1.653777  
hip-hop    91.131595     -72.467678  
metal     38.728342      10.558677  
rock      9.889063      16.868840
```

 *Miró, these coefficients look an awful lot like the coefficients from my interaction analysis, don't they?*

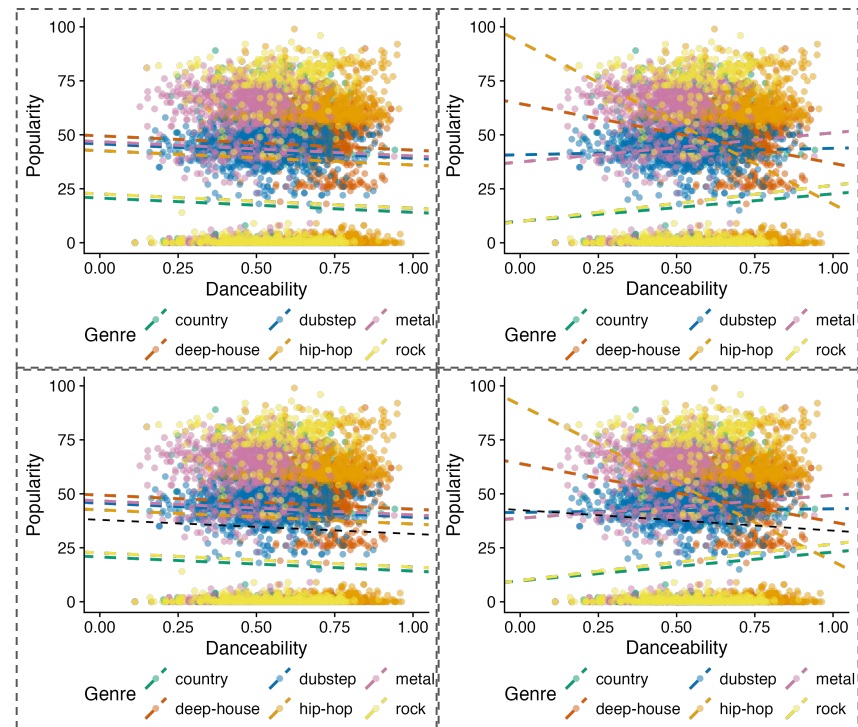
track_genre	danceability.trend	SE	df	lower.CL	upper.CL
country	12.7	6.71	5988	-0.455	25.8
deep-house	-27.6	9.02	5988	-45.275	-9.9
dubstep	3.0	6.59	5988	-9.921	15.9
hip-hop	-75.1	7.01	5988	-88.865	-61.4
metal	13.5	7.20	5988	-0.619	27.6
rock	16.7	6.15	5988	4.653	28.8

Confidence level used: 0.95

Yes, although there is some shrinkage, can you see why? 

Test choice

	Genre shifts the level of popularity	Genre also changes the effect of danceability
These specific genres are of interest (fixed)	Covariate	Moderator
Genres are a sample of groups (random)	Random intercept	Random intercept/slope



Fixed vs. random

Covariate or moderator

💡 Separate level and/or relation for each genre

- ✓ Dependence within genres*
- ✓ Estimates for genres
- ✓ Tests comparing genres
- ✓ Few genres
- ✗ Many genres
- ✗ Few or varying observations per genre
- ✗ Genre-level predictors
- ✗ Generalization to other genres

🦸 *You called me naive? Seems like a good choice after all.*

Random intercept and/or slope

💡 Average level/relation, variation across genre

- ✓ Dependence within genres*
- ✓ Estimates for genres (shrunk)
- ✗ Tests comparing genres
- ✗ Few genres
- ✓ Many genres
- ✓ Few or varying observations per genre
- ✓ Genre-level predictors
- ✓ Generalization to other genres

Ignorance is bliss? 🦹

* If dependence is correctly captured in level and/or relation

Pooling recap

	Intercept	Slope
Simple regression	Complete pooling	Complete pooling
Covariate	No pooling	Complete pooling
Moderation	No pooling	No pooling
Random intercept	Partial pooling	Complete pooling
Random intercept/slope	Partial pooling	Partial pooling

Evaluation | Model comparisons

```
fit_0 <- lm(popularity ~ danceability, data =
spotify_by_genre)
fit_1 <- lmer(popularity ~ danceability + (1 |
track_genre), data = spotify_by_genre)
fit_2 <- lmer(popularity ~ danceability + (1 +
danceability | track_genre), data =
spotify_by_genre)

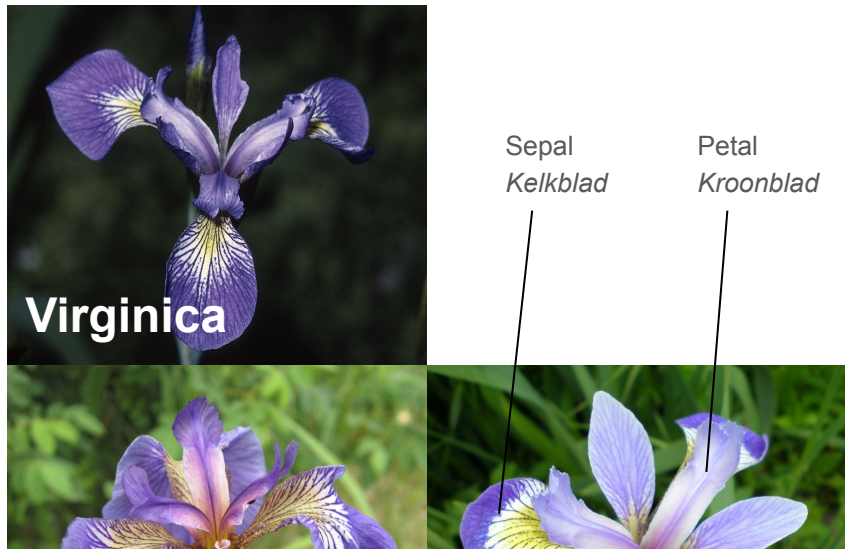
anova(fit_1, fit_2, fit_0)
```

```
Data: spotify_by_genre
Models:
fit_0: popularity ~ danceability
fit_1: popularity ~ danceability + (1 | track_genre)
fit_2: popularity ~ danceability + (1 + danceability | track_genre)
      npar   AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
fit_0     3 57469 57489 -28731    57463
fit_1     4 56468 56495 -28230    56460 1002.92  1 < 2.2e-16 ***
fit_2     6 56362 56403 -28175    56350  109.31  2 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

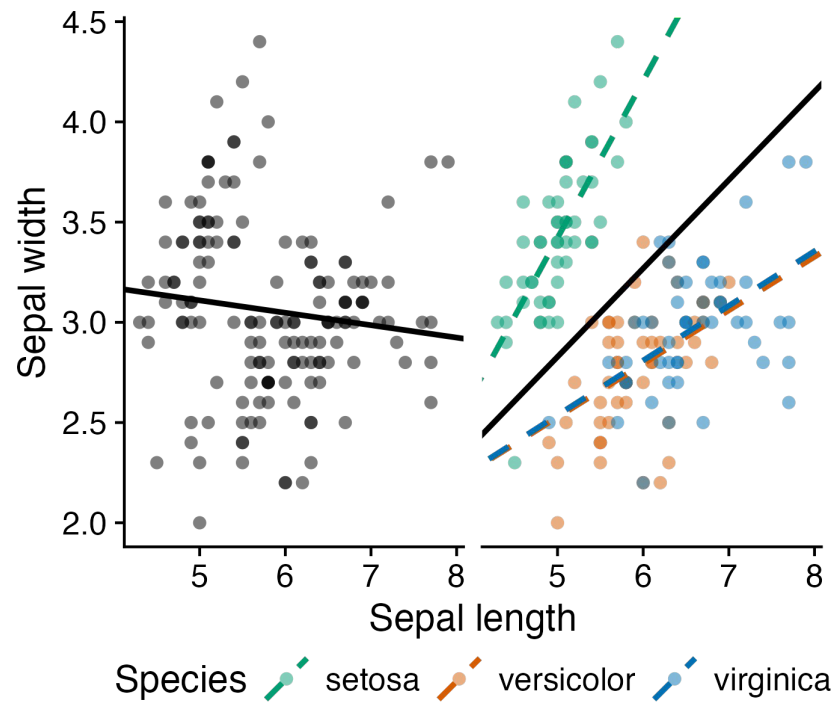
- ❑ What's the best fitting model?
- ❑ Why?

Across genres, what do the relationships between danceability and popularity have in common (fixed) and what sets them apart (random). Evidently significant, because there's a lot that sets them apart: good reason to adjust the average relationship by their genre-dependencies. But, is the average relationship meaningful if there's so much that sets them apart?

Iris



```
fit_ordinary <- lm(Sepal.Width ~ Sepal.Length, data = iris)
fit_random <- lmer(Sepal.Width ~ Sepal.Length + (1 + Sepal.Length | Species), data = iris)
```

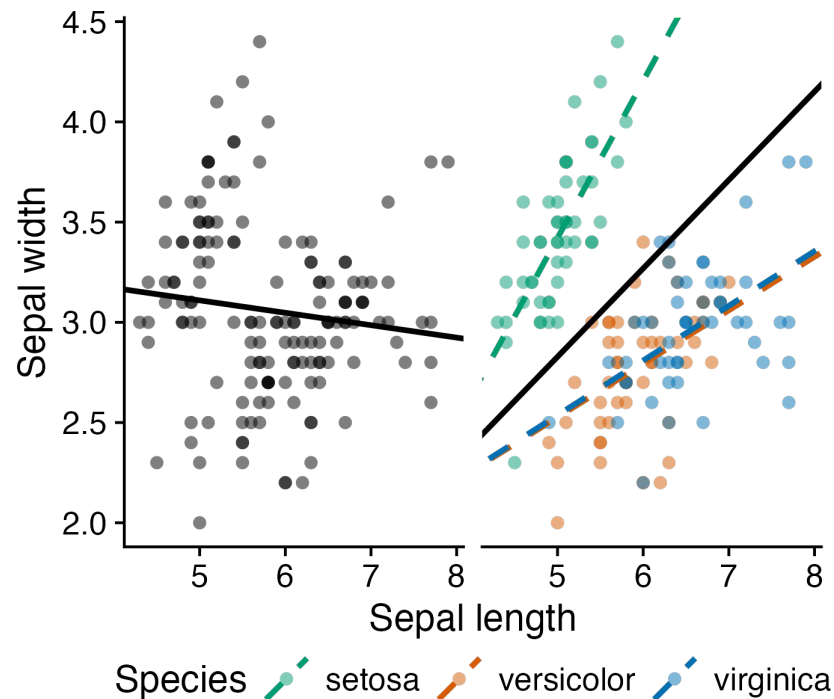


Simpson's paradox

“Simpson's paradox is a phenomenon in probability and statistics in which a trend appears in several groups of data but disappears or reverses when the groups are combined.”

— [Wikipedia](#)

```
fit_ordinary <- lm(Sepal.Width ~ Sepal.Length, data = iris)
fit_random <- lmer(Sepal.Width ~ Sepal.Length + (1 + Sepal.Length | Species), data = iris)
```





Cooling Down



Takeaways

If you hate statistics, choose

- ☐ Complete pooling
- ☐ No-pooling
- ☐ Partial pooling
- ☐ Carpooling





Colophon

Slides

alexandersavi.nl/teaching/

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